Medication adherence is a critical factor in managing chronic illnesses, yet many patients forget to take their prescribed doses on time. This can lead to serious health complications, reduced treatment effectiveness and increased medical costs. Traditional pillboxes rely on manual tracking, which can be ineffective with memory challenges or busy schedules.

This project aims to develop a smart pillbox using an Arduino and an MC-38 magnetic sensor to monitor whether the pillbox has been opened within a specified timeframe. If the pillbox remains unopened past the scheduled medication time, the system will trigger an alert (such as a buzzer, LED, or mobile notification) to remind the user to take their medication. By automating adherence tracking, this smart pillbox can help patients manage their medication schedules more effectively, improving health outcomes and reducing the risk of missed doses.

1 and 3 For our brainstorming, we wanted people to have easy access to this device for those on a tight budget. Almost everyone, no matter their class in society, has medications that they take every day. Since medications are so expensive nowadays, we wanted our pill box to be cheaper. During the process, make sure you plan everything accordingly, have enough time to complete the project, compile a list of items you need to do, test the code, and make sure everyone knows their items and objectives they have to complete. You will have to make multiple prototypes and may have to change different parts of your process, but always make sure to make notes on any that you make, big or small.

2 what went well is that our code worked with the hinge sensor, but or prototype was too big to be a pill box, and we didn’t have time to complete the project to a 100%